Jiajun Zhang

(438) 351-3339 • zhangjohnson729@gmail.com • https://johanssonrein.github.io/

EDUCATION

Bachelor of Science
McGill University 2022-Present
Major: Honors Applied mathematics
Graduation Time: June 2026

RESEARCH INTERESTS

Mathematical Statistics, Biostatistics, Machine Learning, Deep Learning, Causal Inference.

MATHEMATICAL COURSEWORK

Probability and Mathematical Statistics; Regression and ANOVA; Deep Learning; Theory of Machine Learning; Differential Equations and Dynamical Systems; Real Analysis and Abstract Algebra; Calculus and Linear Algebra

RESEARCH EXPERIENCE

Honors Research Project
McGill University 2025.2 - 2025.8

- Research Title: Analyzing Incident and Prevalent Cohort Survival Data
- Research Supervisor: Professor Masoud Asgharian
- Written Report: *An Introduction to Survival Analysis*

This research provides a review of techniques in analyzing right-censored, left-truncated cohort survival data. Approaches include parametric model using likelihood inference; non-parametric model using Kaplan-Meier, Nelson-Aalen and multi-state Johansen-Aalen estimators and their properties; Cox's semi-parametric hazard regression model; Non-parametric regression and hypothesis testing models.

UNDERGRADUATE PROJECTS

- **Deep Learning and Nreural Networks** 2025.11 - 2025.12
  - Project Report: VC Dimension, Prbabilistic Bounds and Neyman-Pearson Classification
  - Project Supervisor: Professor Masoud Asgharian

  This projects studies the theory of Vapnik-Chervonenkis dimension through a probabilistic point of view, and we study Neyman-Pearson classification and its applications.
- **High-Dimensional Regression** 2025.10 - 2025.12
  - Project Report: A Survey on High-Dimensional Regression
  - Project Supervisor: Professor Mehdi Dagdoug

This project studies different regularization methods for high-dimensional regression and the theory of variable selections using Lasso.

- **Dynamical Systems** 2025.3 - 2025.4
  - Project Report: Introduction to Circular Restricted Three-Body Problem
  - Project Supervisor: Professor Antony R. Humphries

This project studies the dynamics of a circular restricted three body problem. Motivated from the closed form solution of a two body problem, an additional massless object is added into the system and derived the set of Lagrange points of the third body as well as their corresponding orbits and stability.

TEACHING AND MENTORING EXPERIENCE

Peer Tutor
McGill University 2023.9 - Present

- Provided individualized help to mentee in mathematics via advice and tutoring.

Undergraduate Mathematics Helpdesk Mentor
McGill University 2024.9 - Present

- Offer open office hours each week to guide students in need and help them with their mathematics coursework.

Undergraduate Course Assistant
McGill University

- Grade student's homework and provide feedback to professors.
  - Course Assistant for MATH 254: Honors Analysis 1 2025.9 - 2025.12
  - Course Assistant for MATH 248: Honors Vector Calculus 2024.9 - 2024.12
  - Course Assistant for MATH 223: Linear Algebra 2023.9 - 2024.4

HONORS AND AWARDS

Tomlinson Undergraduate Award 2025.3 and 2024.10
McGill University

- Awarded to undergraduate course assistants.

VOLUNTEERING EXPERIENCE

Math & Stats Buddy Mentorship Program 2025.9 - Present
McGill University

- This program matches upper year undergraduate students from math department with freshman students, where mentor will provide advice and help for mentee.

McGill Robotics Club 2023.9 - 2024.4
McGill University

- Duty: Working with Drone Group as a software developer on drone's vision system.

EXTRACURRICULAR PROJECTS

Math Resources Website 2024 - Present

- Website link: Math Resources
- Developed a website to provide free math resources for undergraduate students and help them learn.

SKILLS

- **Technical Skills:** LaTeX, Github, R, Python, Java

LANGUAGES

Mandarin (Native), English (Proficient), French (Intermediate)